NAME: _____ CLASS: _____ TEACHER: _____

**KEY TOPIC 3** CHRONOLOGICAL ENQUIRY SEQUENCE • C1700–C1900

# Medicine in 18th & 19th Century Britain

*c1700–c1900: Ideas on Causes, Prevention, Hospital Care, Surgery & Public Health*




HISTORICAL PRIMARY RECORD • 1802 — HISTORICAL SATIRE ARCHIVE

**James Gillray, The Cow-Pock—or—the Wonderful Effects of the New Inoculation! (1802)**

Hand-coloured etching lampooning early Victorian fears that Edward Jenner's cowpox vaccine would cause patients to sprout bovine horns and calves.

PRIMARY VISUAL EVIDENCE — THE BRITISH MUSEUM DEPARTMENT OF PRINTS & DRAWINGS

**PEARSON EDEXCEL** OPTION 11 • KEY TOPIC 3 CONTENT — OFFICIAL SPECIFICATION DIRECTORY

## 1 Ideas About the Cause of Disease and Illness

- Continuity and change in explanations of the cause of disease and illness. The influence in Britain of **Pasteur's Germ Theory** and **Koch's work on microbes**.

🎯 **CAUSATION TURNING POINT**
**From Spontaneous Generation to Germs:** How Pasteur's 1861 swan-neck flask experiments and Koch's chemical dye staining isolated specific pathogens, decisively destroying miasma.

## 2 Approaches to Prevention and Treatment

- The extent of change in care and treatment: improvements in hospital care and the **influence of Nightingale** on nursing and hospitals in Britain. The impact of **anaesthetics and antiseptics** on surgery.
- New approaches to prevention: the development and use of vaccinations and the **Public Health Act (1875)**.

## 3 Case Studies

- Key individual: **Jenner and the development of vaccination**.
- Fighting **Cholera in London (1854)**; attempts to prevent its spread; the significance of **Snow and the Broad Street pump**.

**EXAM MATRIX** EDEXCEL PAPER 1 SECTION B EXAMINATION TARGETING FRAMEWORK • THEMATIC STUDY [36M + 4M SPAG • 55 MINS] — SPECIFICATION 1HI0/11

**Q3 • 4 MARKS** — AO1/AO2 CONCEPT DRILL

### Similarity or Difference Across Eras

**Target:** Explain 1 way [ideas / treatments / care] in this era were similar or different to another period.
**Rule:** Provide specific factual evidence from *both* named periods to secure 4/4 marks (approx. 5 mins).

Practice: In-Workbook 4-Mark Comparative Tasks

**Q4 • 12 MARKS** — AO1/AO2 CAUSATION

### Explain Why [Causal Change] Occurred

**Target:** Multi-causal analytical essay explaining why medical change or continuity took place.
**Rule:** Write 3 developed PEEL paragraphs. Must address both stimulus points + 1 distinct point of own knowledge (18 mins).

Practice: Chronological Inquiry Spine & Causation Drills

**Q5/6 • 16+4 SPaG** — AO1/AO2/AO4 SYNOPTIC

### 'How Far Do You Agree?' Evaluative Essay

**Target:** Choose Q5 OR Q6. Evaluative thematic essay assessing factor significance or change over time.
**Rule:** Balanced argument (for/against) with sustained criteria and a decisive historical verdict (30 mins).

Practice: Full 16-Mark Extended Writing Frames

# Living Timeline • Part 1: Vaccines, Anaesthetics & Early Sanitation (1700–1854)



**Living Timeline Dual-Coding Protocol:** Complete each milestone sketchpad using dual-coding (combining visual storytelling symbols, causal arrows, and dates) to master the chronological spine across the centre-fold.

### 1796 • Edward Jenner: Smallpox Vaccination & Immunology

**Key Topic 3.2**

Jenner inoculates James Phipps with cowpox pus from Sarah Nelmes, proving cowpox confers immunity against smallpox. Publishes findings in 1798. Despite initial Royal Society scepticism, vaccination eradicates inoculator infection risks and founds immunology.

### 1847 • James Young Simpson: Chloroform & Anaesthesia

**Key Topic 3.4**

Simpson discovers the potent anaesthetic properties of chloroform in Edinburgh, ending agony during surgery. Queen Victoria's use of chloroform during childbirth (1853) popularises it, enabling complex operations despite the initial 'Black Period' of surgery.

### 1848 • First Public Health Act & The General Board of Health

**Key Topic 3.5**

Spurred by Chadwick's 1842 report and epidemic cholera, Parliament passes the permissive 1848 Act. Towns can create local boards of health to supply clean water and drainage, though lack of compulsion leaves most cities vulnerable to filth and contagion.

### 1854 • John Snow: Broad Street Pump & Waterborne Cholera

**Key Topic 3.5**

During the Soho epidemic, Snow compiles his famous spot map linking 500+ fatal cholera cases to the Broad Street pump. Removing the handle immediately halts the outbreak, mathematically disproving the miasma theory seven years before Pasteur's Germ Theory.



*The 18th century relied on empirical observation; the 19th century unlocked the laboratory.*

# Living Timeline • Part 2: Nursing, Microbes & Antiseptics (1854–1875)



**Living Timeline Dual-Coding Protocol:** Draw visual symbols for Crimean hospital sanitation, swan-neck flasks, Lister carbolic sprays, and municipal sewers across the timeline below.

### 1854–56 • Florence Nightingale: Hospital Sanitation at Scutari

**Key Topic 3.3**

During the Crimean War, Nightingale enforces rigorous cleanliness, ventilation, and fresh nutrition at Scutari hospital, slashing soldier death rates from 40% to 2%. Returns to Britain to establish the Nightingale Training School and author Notes on Nursing (1859).

### 1861 • Louis Pasteur: Germ Theory & Swan-Neck Flask Experiments

**Key Topic 3.1**

Commissioned by Lille brewers, Pasteur disproves spontaneous generation using swan-neck flasks, demonstrating that airborne microbes cause fermentation and decay. In 1861, he publishes Germ Theory, proving microscopic germs cause specific diseases in living organisms.

### 1865–67 • Joseph Lister: Carbolic Acid Spray & Antiseptic Surgery

**Key Topic 3.4**

Applying Pasteur's Germ Theory to wound sepsis, Lister uses carbolic acid sprays, dressings, and hand-washing during surgery at Glasgow Infirmary. His antiseptic techniques reduce amputee mortality from 46% to 15%, paving the way for modern aseptic operating theatres.

### 1875 • The Second Public Health Act: End of Laissez-Faire

**Key Topic 3.5**

The 1875 Public Health Act ends laissez-faire by making public health compulsory. Municipal councils are legally required to provide clean piped water, subterranean sewer drainage, street lighting, and medical officers of health, transforming urban life expectancy.

# Why did it take 30 years for Germ Theory to transform medical understanding in Britain?

AUDIO • QUIZ

## 1 c1700–1850s • Why did doctors believe disease created germs, rather than vice versa?

- Spontaneous generation theory
- Miasma stench blamed
- Pre-bacteriology limits

## 2 1861 • How did the swan-neck flask disprove spontaneous generation?

- Louis Pasteur (1861)
- Swan-neck flask test
- Airborne microbes proved

## 3 1867–1876 • Why did British doctors refuse to believe germs caused human illness?

- British medical scepticism
- John Tyndall experiments
- Charlton Bastian opposition

## 4 1876–1882 • How did Koch's staining and culture methods prove specificity?

- Robert Koch bacteriology
- Agar jelly plates
- Methyl violet stains

## 5 1883–1890 • Why was photomicrography crucial in winning British doctors over?

- Tuberculosis bacteria (1882)
- Cholera microbe (1883)
- High-resolution microphotography



*Pasteur proved that rotting soup doesn't spontaneously create germs; Koch proved which specific germs caused tuberculosis.*

## Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

**Explain why there was rapid progress in understanding the causes of disease in the period c1860–c1890. [12 marks]**

**Stimulus:** **Louis Pasteur's Germ Theory (1861)** **Robert Koch's work on bacteria** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 3.1). In the drawing box, sketch Pasteur's swan-neck flask experiment and Koch's agar plate. Annotate how identifying specific bacteria ended miasma theory forever! **[ ] Done**

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which ideas about the cause of disease in the late nineteenth century (c1860–c1890) were different from ideas in the Renaissance (c1500–c1700). [4 marks]**

*Contrast laboratory discoveries of specific microbes with earlier beliefs in miasma and spontaneous generation.*

# Why did Edward Jenner face fierce resistance despite discovering the smallpox vaccine?

AUDIO • QUIZ

## 1 1721–1795 • Why was inoculation dangerous for poor urban communities?

- Smallpox variolation inoculation
- Suttonian method fees
- Infection risk remains

## 2 1796–1798 • How did Jenner use scientific experimentation to test folklore?

- Edward Jenner (1796)
- James Phipps vaccination
- Cowpox blister pus

## 3 1798–1810s • Why did London doctors and church leaders resist vaccination?

- Royal Society scepticism
- Anti-vaccine cartoon satire
- Cowpox contamination fears

## 4 1802–1840 • Why did the British government abandon laissez-faire for vaccination?

- Parliamentary grants (£30,000)
- 1840 Vaccination Act
- Inoculation banned legally

## 5 1852–1871 • Why did making vaccination compulsory spark fierce public protests?

- Compulsory Vaccination (1853)
- 1871 Enforcement Act
- Dramatic smallpox eradication

*Cartoons showed vaccinated people growing cow heads and horns: proof that anti-vax memes existed long before social media.*

## Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

**'Edward Jenner's development of the smallpox vaccine was the most significant breakthrough in the prevention of disease in the period c1700–c1900.' How far do you agree? [16+4 marks]**

**Stimulus:** **Inoculation** **The Public Health Act (1875)** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *On the one hand, a major breakthrough was... • For example, ... • This was significant because... • On the other hand, an alternative factor was... • In particular, ... • Overall, the most decisive factor was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 3.2). In the drawing box, sketch Jenner inoculating James Phipps with cowpox. Annotate how this empirical experiment laid the foundation for modern immunology! **[ ] Done**



*Jenner saved millions with cowpox, yet died before anyone could explain what a virus actually was.*

*Continue your response to Question 5/6 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which methods of preventing smallpox in the late eighteenth century were different from methods used in the seventeenth century. [4 marks]**

*Contrast safe cowpox vaccination with earlier reliance on prayer, quarantine, and risky variolation.*

# How did Florence Nightingale transform hospitals from death traps into places of cure?

AUDIO • QUIZ

## 1 c1700–1850 • Why were early Victorian hospital wards notorious death traps?

- Pauper hospital squalor
- Untrained nurse reputation
- Cross-infection septic wards

## 2 1854–1856 • How did Nightingale's sanitary reforms drastically cut the death rate?

- Crimean War (1854)
- Scutari hospital sanitation
- Death rate plummeted (42% to 2%)

## 3 1859 • Why did Nightingale's belief in miasma still produce effective hygiene?

- Notes on Nursing (1859)
- Sanitary statistical charts
- Pavilion hospital architecture

## 4 1860s • How did the pavilion plan prevent cross-infection between wards?

- Separate pavilion wards
- Cross-ventilation high windows
- Tiled washable surfaces

## 5 1860–1890s • How did nurse training permanently raise the standard of hospital care?

- St Thomas' Training School
- Nightingale Fund (£44,000)
- Professional disciplined matrons



*Nightingale believed miasma caused disease, yet her obsession with fresh air, scrubbing, and clean drains saved thousands of lives.*

## Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

**Explain why the standard of hospital care and nursing improved so significantly in the second half of the nineteenth century. [12 marks]**

**Stimulus:** **Florence Nightingale's work in the Crimea (1854–56)** | **The pavilion hospital plan** | *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 3.3). In the drawing box, sketch the Pavilion Hospital plan with its separate, ventilated pavilions. Annotate how cross-ventilation prevented hospital gangrene! **[ ] Done**

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which hospital care in the late nineteenth century was different from hospital care in the Medieval period. [4 marks]**

*Contrast modern sanitary pavilion wards and trained medical care with medieval monastic care.*

# Why did the conquest of pain in surgery temporarily increase the patient death rate?

AUDIO • QUIZ

## 1 c1700–1846 • Why was surgical speed essential before the discovery of anaesthetics?

- Speed-focused surgery pain
- Surgical shock mortality
- Ether & nitrous oxide

## 2 1847–1853 • Why did the death of Hannah Greener cause panic among surgeons?

- James Simpson (1847)
- Chloroform anaesthetic vapour
- Queen Victoria childbirth (1853)

## 3 1846–1865 • Why did conquering surgical pain lead to higher patient death rates?

- 'Black Period' surgery
- Deeper invasive operations
- Hospital gangrene surge

## 4 1865–1870s • How did Lister apply Germ Theory to solve hospital gangrene?

- Joseph Lister (1865)
- Carbolic acid spray
- Compound fracture antisepsis

## 5 1880s–1890s • What is the crucial difference between antiseptic and aseptic surgery?

- Aseptic surgical theatre
- Autoclave steam sterilisation
- Sterile gloves & masks

*Chloroform conquered pain, but Lister conquered the invisible enemy: the rotting microbes thriving on surgeons' unwashed hands.*

## Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

**Explain why surgery became significantly safer in the period c1840–c1900. [12 marks]**

**Stimulus:** **James Simpson's discovery of chloroform (1847)** | **Joseph Lister's use of carbolic acid (1865)** *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 3.4). In the drawing box, sketch Lister's carbolic donkey engine spraying mist over an open wound. Annotate the difference between antiseptic and aseptic surgery! **[ ] Done**



*Surgeons used to take pride in blood-caked frock coats; Lister forced them to wash their hands and operate in carbolic acid mist.*

*Continue your response to Question 4 below:*

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which surgical operations in the early nineteenth century (before 1846) were similar to surgery in the Medieval period. [4 marks]**

*Focus on raw surgical speed and physical restraint before effective chemical anaesthetics were discovered.*

# Why did it take 20 years for John Snow's cholera discovery to change government policy?

AUDIO • QUIZ

## 1 1831–1848 • Why did towns refuse to set up health boards under the 1848 Act?

- Cholera epidemics (1831–1854)
- Laissez-faire government policy
- Permissive 1848 Health Act

## 2 1854 • How did Snow use epidemiological mapping to prove water transmission?

- John Snow Soho (1854)
- Broad Street spot map
- Pump handle removed

## 3 1854–1858 • Why did the medical establishment refuse to accept Snow's water theory?

- Water company comparison
- Southwark vs Lambeth
- Waterborne transmission proved

## 4 1858–1865 • Why did fear of miasma paradoxically save thousands from cholera?

- Great Stink (1858)
- Joseph Bazalgette sewers
- Underground London interceptors

## 5 1875 • Why was the 1875 Act the definitive death blow to laissez-faire?

- 1875 Public Health Act
- Compulsory town sewers
- End of laissez-faire



*Snow removed the Broad Street pump handle in 1854, proving cholera was water-borne seven years before Germ Theory.*

## Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

**'The work of John Snow was the main reason for improvements in public health in the nineteenth century.' How far do you agree? [16+4 marks]**

**Stimulus:** Edwin Chadwick's 1842 Report | The Public Health Act (1875) *(You must also use information of your own)*

**ANALYTICAL CONNECTIVES:** *On the one hand, a major breakthrough was... • For example, ... • This was significant because... • On the other hand, an alternative factor was... • In particular, ... • Overall, the most decisive factor was... because...*

***Write your response in the space provided below:*** *(Response continues on facing page)*

**TIMELINE MISSION** Turn to Pages 2–3 (Key Topic 3.5). In the drawing box, sketch John Snow's Broad Street pump map and Bazalgette's sewer network. Annotate how cholera forced Parliament to abandon laissez-faire! [ ] Done

*Continue your response to Question 5/6 below:*

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [ / 4 ] • 5 MINS**

**Explain one way in which government action on public health in the late nineteenth century was different from government action in the Renaissance. [4 marks]**

*Contrast compulsory public health legislation with earlier temporary emergency measures.*

## KEY TOPIC 3 NOTES & EXAMINATION STRATEGY

*Notes: Record key causal links, essay plans, and teacher feedback for this unit.*

# STUDENT ASSESSMENT RECORD • KEY TOPIC 3 TRACKER

Paper 1: Medicine in Britain, c1250–present • 18th & 19th Century Britain (c1700–c1900)

**PUPIL:** _____ **TARGET GRADE:** ☐ **PREDICTED:** ☐ **ATTITUDE:** 1 • 2 • 3 • 4 • 5

| # | ENQUIRY / SPECIFICATION FOCUS | DO NOW (10M) | Q3 SIM/DIFF (4M) | Q4/5 EXTENDED | LESSON TOTAL |
|---|---|---|---|---|---|
| 1 | **KT3.1: Ideas on Causes: Pasteur's Germ Theory & Koch's Bacteriology**<br>*"Why did it take 30 years for Germ Theory to transform medical understanding in Britain?"* | / 10 | / 4 | Q4 Explain: / 12 | / 26 |
| 2 | **KT3.2: Approaches to Prevention: Edward Jenner & The Smallpox Vaccine**<br>*"Why did Edward Jenner face fierce resistance despite discovering the smallpox vaccine?"* | / 10 | / 4 | Q4 Explain: / 12 | / 26 |
| 3 | **KT3.3: Improvements in Hospital Care: Florence Nightingale & Professional Nursing**<br>*"How did Florence Nightingale transform hospitals from death traps into places of cure?"* | / 10 | / 4 | Q4 Explain: / 12 | / 26 |
| 4 | **KT3.4: The Surgical Revolution: Simpson's Chloroform & Lister's Antiseptics**<br>*"Why did the conquest of pain in surgery temporarily increase the patient death rate?"* | / 10 | / 4 | Q4 Explain: / 12 | / 26 |
| 5 | **KT3.5: Public Health & Cholera: John Snow & The 1875 Public Health Act**<br>*"Why did it take 20 years for John Snow's cholera discovery to change government policy?"* | / 10 | / 4 | Q5/6 Essay: / 20 | / 34 |
| **KEY TOPIC 3 CUMULATIVE ASSESSMENT TOTALS** | | **Do Now: / 50** | **Q3: / 20** | **Ext: / 68** | **/ 138** |

## TEACHER FORMATIVE ASSESSMENT • WWW / EBI FEEDBACK

**WHAT WENT WELL (WWW):**

_____

_____

_____

**EVEN BETTER IF (EBI):**

_____

_____

_____

**Teacher Signature:** _____ **Date:** _____

## DIGITAL QUIZZING HUB • 20-QUESTION PRACTICE QUIZZES

Scan with smartphone camera to open live interactive 20-question self-marking quizzes

| KT 3.1 | KT 3.2 | KT 3.3 | KT 3.4 | KT 3.5 |
|---|---|---|---|---|
| SCAN TO QUIZ | SCAN TO QUIZ | SCAN TO QUIZ | SCAN TO QUIZ | SCAN TO QUIZ |
| **Score: / 20** | **Score: / 20** | **Score: / 20** | **Score: / 20** | **Score: / 20** |



*Retain this student assessment record for parent consultations and GCSE revision monitoring.*